

Telemetry-Driven Scale-Across over On-Demand Optical Slices in Federated Data Centres

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Abstract We demonstrate telemetry-driven scale-across of distributed workloads over on-demand optical slices between geo-distributed data centres on a disaggregated multi-vendor testbed. Analytics anticipate SLA pressure across compute- and network-bound phases and trigger QoT-validated slice adaptation. ©2026 The Author(s)

Introduction

Modern hyper-distributed applications increasingly stress individual data centres (DCs), while data-locality, sovereignty and latency requirements prevent all components from being consolidated at a single site. A workload that initially fits within one DC may therefore need to scale-across geo-distributed DCs without violating its service-level agreement (SLA). Preserving the SLA requires reassigning parts of the workload to additional DCs and establishing inter-DC connectivity over high-capacity, low-latency optical paths, typically realized through programmable IP-over-WDM (IPoWDM) where coherent interfaces are integrated with packet-layer equipment over a shared optical transport infrastructure.

Programmable optical transport exposes this connectivity as an abstracted, multi-tenant service through interfaces such as TAPI within an ACTN framework [1], realizing the spectrum-as-a-service paradigm in which per-tenant spectrum is allocated on demand over a shared, partially disaggregated infrastructure. Combined with transport network slicing [2] and QoT digital twins [3], this lets tenants request QoT-assured optical slices. In current practice, however, such slices are provisioned independently of the run-time state of the applications they support, dimensioned once at request time and held flat for the service lifetime even when the workload is non-stationary. Many distributed workloads make this gap visible, alternating between compute-bound and network-bound phases in a regular pattern of periodic synchronisation and aggregation steps that concentrate traffic on the inter-DC link, interleaved with longer local-computation intervals, while request-driven services alternate between bursts and idle windows.

Extending our zero-touch intent-driven deployment demonstration [4] from one-shot placement to run-time scale-across, we close the loop between live compute telemetry, the workload's internal phase structure, and the on-demand pro-

visioning and dynamic adaptation of the corresponding inter-DC optical slice. We introduce two control-plane components, a Telemetry Engine (TE) that ingests per-pod compute, traffic and, where available, framework-level signals from the federated clusters, and an Analytics Engine (AE) that consumes the TE stream together with transport telemetry and drives both control loops. A workload initially served entirely at one data centre (DC1) is monitored by the TE, while the AE anticipates SLA issues and emits structured events to the Service Orchestrator (SO). When the AE predicts that DC1 will approach saturation, the SO requests a placement update from the Holistic Resource Allocation Tool (H-RAT), which jointly optimises compute placement and an inter-DC optical slice through the transport control plane.

The slice is provisioned by TeraFlowSDN, Flex-Opt and ONOS [5, 6] and QoT-validated by a GNPpy digital twin, after which the offloaded microservices are migrated to a second data centre (DC2) and interconnected over the slice. Once active, the slice is adapted by the AE to track the running workload's phases, widened ahead of network-heavy bursts and shrunk between them, extending spectrum-as-a-service from static per-tenant allocation to dynamic, workload-driven reallocation. The demonstration runs two workloads with distinct phase structures. The first is a distributed iterative compute job across DC1 and DC2, each iteration alternating local computation with an inter-DC synchronisation step. The second is a multi-component request-driven service with a burst structure. The complete loop is zero-touch, with live GUIs reporting scale-across time, slice-establishment time, inter-DC round-trip time and bandwidth, and QoT margin.

System Architecture

Fig. 1 shows the demonstration setup. The optical infrastructure is based on a core optical network testbed hosted at Links Foundation labs in Turin. The architecture couples a federated com-

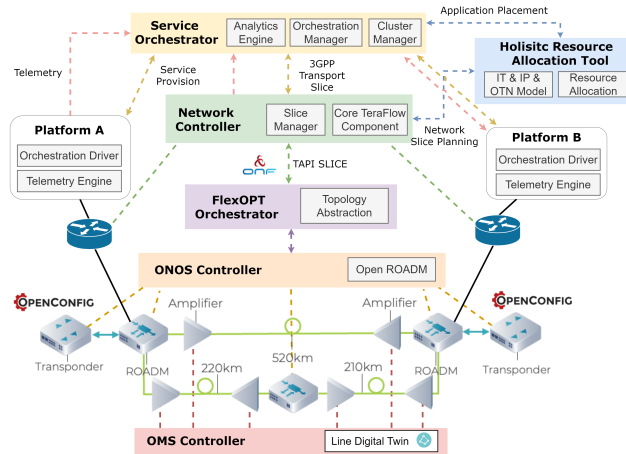


Fig. 1: Prototype architecture for telemetry-driven scale-across of distributed workloads over on-demand optical slices.

pute domain with a programmable optical transport domain through the telemetry-driven control loop introduced in the previous section.

Compute domain. The federated compute resources comprise DC1 and DC2, each supported by a small K8s cluster or a single worker station. A dedicated Orchestrator Driver in each site registers its worker nodes and resources with the SO and applies the deployment manifests it receives. Because it handles container deployment locally, upstream components see a uniform federation whether a site is a multi-node K8s cluster or a single workstation. The TE collects live per-pod compute and traffic metrics from both sites, CPU, memory, request rate and inter-component latency, and, where available, framework-level signals from the running workload. The AE consumes the TE stream together with transport telemetry, detects the workload's current phase and anticipates SLA pressure, while the SO consumes AE events to decide when and how to scale across DCs and H-RAT performs the joint compute-network optimisation that yields the new placement.

Transport domain. Inter-DC connectivity is delivered over a disaggregated, multi-vendor optical line system, a three-ROADM chain with EDFAs, under a hierarchical control plane. ONOS programs the ROADM and OMS devices and exposes device and link telemetry through OpenROADM and OpenConfig models [6]. FlexOpt virtualises the line system and, through TAPI, presents each tenant with an abstracted topology over which spectrum and connectivity services can be requested, while the operator retains device-level optical state [1, 5]. TeraFlowSDN acts as the slicing controller, enforcing tenant isolation and IPoWDM control, computing feasible media channels, and leveraging a GNPpy digital twin for QoT-safe provisioning and margin validation [2, 3]. The DC owner drives its own coherent IPoWDM interfaces at the cluster gateways and requests inter-DC connectivity through this abstracted TAPI/ACTN interface.

Cross-layer closed loop. The distinguishing feature of the architecture is the run-time coupling of the two domains through the TE-AE integration. At each control interval the AE evaluates whether the active placement will continue to meet the SLA under current and predicted telemetry. A first loop reacts to resource saturation, emitting a scale-across event to the SO, which requests a new placement from H-RAT together with the inter-DC connectivity requirement (bandwidth and latency), and an inter-DC slice through the transport control plane. Once the slice is established and QoT-validated, its service endpoints are returned to the SO, which finalises the deployment of the off-loaded components in the selected DC and steers traffic between them over the slice. A second loop, active once the workload runs across both sites, tracks its alternating computing and networking phases and instructs the TeraFlowSDN Network Controller (NC) to widen or shrink the active slice so that allocated spectrum follows demand. The two loops together form a single, zero-touch cross-layer control surface spanning the compute and optical layers, driven by shared telemetry.

Demonstration Workflow

The demonstration runs zero-touch across both domains in three phases that expose the AE's two control loops sequentially, with live KPIs on web-based UIs. **Phase I: Single-site operation and monitoring.** The request-driven workload is deployed entirely in one DC, which initially has sufficient capacity, so no inter-DC slice is required. While the application is served, the TE streams per-container utilisation, request rate, and inter-component latency to the AE, and the UIs display the resulting state and the AE's running margin against the SLA, establishing a clean baseline before any cross-layer action.

Phase II: Telemetry-driven scale-across and inter-DC slicing. A controlled load increase is applied to the request-driven workload, driving the selected DC to saturation. The workflow (Fig. 2a)

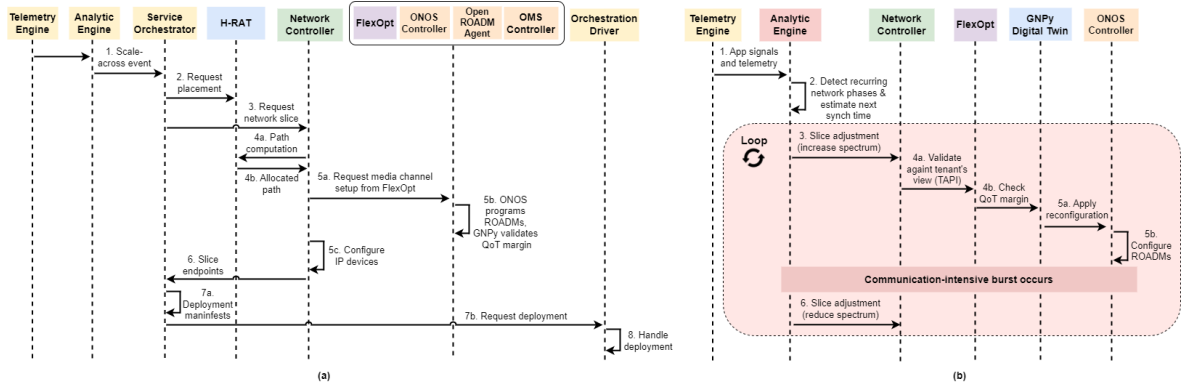


Fig. 2: Demonstration workflows. (a) Telemetry-driven scale-across and inter-DC slice establishment in Phase II. (b) Workload-phase-aware dynamic slice adaptation in Phase III.

then proceeds as follows. **(1)** The AE, consuming the TE stream, anticipates an SLA breach and emits a scale-across event to the SO. **(2)** The SO asks H-RAT for a new placement that offloads compute-intensive microservices to the other DC, together with the inter-DC connectivity requirement between the two parts of the application. **(3)** The SO requests an inter-DC optical slice from the transport control plane over the TAPI/ACTN interface with the required QoS. **(4)** The NC Slice Manager evaluates the request and computes feasible optical paths. **(5)** The tenant's IPoWDM control requests media-channel set-up from FlexOpt, which uses ONOS to configure the channels; ONOS programs the ROADMs via OpenROADM agents, and the GNPY digital twin validates QoT and optical margin before activation. **(6)** Once the slice is active, the transport control plane returns the service-endpoint pair to the SO. **(7)** The SO generates the updated deployment manifests and instructs the Orchestration Drivers to manage the workload migration. **(8)** Inter-service endpoints are reconfigured to use the slice endpoints, so traffic between DCs traverses the QoT-validated slice and the workload runs across the two sites with service continuity preserved.

Phase III: Workload-phase-aware dynamic slice adaptation. With the federation now active, the demo switches to a workload whose computing and networking phases alternate regularly, here a small distributed iterative compute job across worker nodes in the two sites, used to exercise the second control loop. Each iteration alternates a local-computation phase with a short, network-heavy synchronisation between DCs, producing a predictable pattern on the inter-DC link that the AE anticipates and tracks. The workflow (Fig. 2b) proceeds as follows. **(1)** The TE forwards framework-level iteration signals from the workload to the AE, alongside the standard per-container compute and traffic metrics. **(2)** The AE recognises the recurring networking phase and estimates the timing of the next synchronisation. **(3)** Ahead of each anticipated aggregation, the AE issues a slice ad-

justment request to NC with the target spectrum width. **(4)** NC validates the request against the tenant's TAPI-abstracted view of the line system through FlexOpt and consults the GNPY digital twin to confirm QoT margin under the new configuration. **(5)** ONOS applies the reconfiguration to the ROADMs via OpenROADM agents, adjusting the allocated spectrum ahead of the networking burst. **(6)** Once the burst completes and the workload returns to its compute phase, the AE issues a complementary adjust request that reduces the slice back.

Live KPIs. Throughout the demonstration, the GUIs expose several live KPIs: (i) the DC1 utilisation trajectory and the scale-across trigger, (ii) the scale-across time decomposed into placement decision, optical-slice establishment (including GNPY QoT validation), and container start-up, (iii) the inter-DC round-trip time and achieved bandwidth over the slice, verified against the target QoS, and (iv) the GNPY-estimated QoT (GSNR) margin together with application-level request continuity.

Conclusion

We demonstrate telemetry-driven scale-across of distributed workloads over geo-distributed data centres, with on-demand provisioning and dynamic adaptation of the inter-DC slice. A dedicated Telemetry Engine and Analytics Engine close the loop between live compute telemetry, the workload's phase structure and the transport control plane, reusing the SO and H-RAT orchestration layer. Two cooperating loops are exercised, a reactive one that scales the workload across DCs and establishes a QoT-validated slice ahead of an SLA breach, and a phase-aware one that resizes the slice in step with its synchronisation pattern. Slice provisioning and reconfiguration use TeraFlowSDN, FlexOpt and ONOS over a multi-vendor disaggregated line, with GNPY validating QoT margin. Live KPIs evidence SLA-preserving operation, extending spectrum-as-a-service from static to dynamic allocation and enabling energy-, cost- and carbon-aware re-optimisation.

Acknowledgements

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References

- [1] Ó. G. de Dios, J. P. F.-P. Giménez, R. Casellas, and L. Roelens, "Demonstration of multi-tenant optical transport network slicing with t-api", in *2024 15th International Conference on Network of the Future (NoF)*, 2024, pp. 154–158. DOI: 10.1109/NoF62948.2024.10741493.
- [2] L. Gifre et al., "Experimental demonstration of transport network slicing with sla using the teraflowsdn controller", in *European Conference on Optical Communication (ECOC) 2022*, Optica Publishing Group, 2022, Tu2.3. DOI: 10.1364/ECEOC.2022.Tu2.3. [Online]. Available: <https://opg.optica.org/abstract.cfm?URI=ECEOC-2022-Tu2.3>.
- [3] R. Ambrosone et al., "Telemetry-driven QoT digital twin for dynamic service provisioning", in *Proc. Optical Network Design and Modelling (ONDM)*, 2025.
- [4] A. Kretsis et al., "Intent-driven slicing across federated compute and optical transport networks", in *Optical Fiber Communication Conference (OFC) 2026*, ser. Technical Digest Series, Optica Publishing Group, 2026, paper M3Z.6.
- [5] R. Casellas et al., "Photonic device programmability for autonomous optical networks", *J. Opt. Commun. Netw.*, vol. 16, no. 8, pp. C1–C12, 2024.
- [6] A. Giorgetti, A. Sgambelluri, F. Paolucci, N. Sambo, P. Castoldi, and F. Cugini, "Control of open and disaggregated transport networks using the open network operating system (ONOS)", *J. Opt. Commun. Netw.*, vol. 12, no. 2, A171–A181, 2020.